Economic And Social Determinants Of Life Expectancy

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Abstract

This study examines the relationship between economic and social determinants and life expectancy across countries using panel data from 142 countries over the period of 14 years from 2010-2023. The analysis focuses on key variables, including GDP per capita, health expenditure, education, and unemployment, to assess their impact on population health outcomes. By employing panel regression techniques, namely pooled OLS, fixed effects, and two-way fixed effects models, the study accounts for both country-specific and time-specific heterogeneity.

The empirical results reveal that GDP per capita and education are strong and consistent determinants of life expectancy, with higher income levels and greater educational attainment significantly improving health outcomes. In contrast, health expenditure shows mixed effects, becoming negative and significant in fixed effects specifications, suggesting that the efficiency of healthcare systems may be more important than the level of spending alone. The impact of unemployment varies across model specifications, reflecting underlying socio-economic complexities.
Furthermore, the analysis incorporates the effects of global shocks by introducing a COVID-19 dummy variable for 2021. The results reveal a statistically significant negative impact on life expectancy, highlighting the adverse consequences of the pandemic on population health.

Overall, the findings emphasise the importance of economic development, human capital formation, and effective healthcare systems in improving life expectancy. The study provides relevant policy implications by suggesting that governments should focus not only on increasing income levels and healthcare expenditure but also on enhancing the efficiency of healthcare delivery and investing in education. While the analysis offers robust evidence, future research could incorporate additional variables and explore non-linear relationships to further understand the determinants of health outcomes.

1. Introduction

Throughout history, there has been a strong and persistent link between economic prosperity and human well-being. In modern economic discourse, income levels are widely recognised as a key determinant of living standards, influencing access to healthcare, education, and overall quality of life. Among the various indicators used to measure well-being, life expectancy stands out as one of the most comprehensive, as it reflects not only the effectiveness of healthcare systems but also broader socio-economic conditions. This motivates the central empirical question: to what extent does economic development translate into longer and healthier lives?

While it may be intuitive to assume that higher income levels lead to better health outcomes, the relationship between economic growth and life expectancy is complex and multifaceted. Life expectancy is shaped by a wide range of factors, including healthcare accessibility, educational attainment, labour market conditions, and institutional environments. As such, focusing solely on income risks overlooking other critical determinants of population health. A more comprehensive analytical framework is therefore required to capture the combined influence of economic and social variables.

This study addresses this gap by examining the impact of GDP per capita, health expenditure, education, and unemployment on life expectancy using a panel dataset of 142 countries over the period 2010-2023. To ensure a robust empirical analysis, panel data techniques are employed, including pooled OLS, fixed effects, and two-way fixed effects models, allowing for the control of both country-specific and time-specific heterogeneity.

In addition, the study incorporates the effects of global shocks, particularly the COVID-19 pandemic, by introducing a dummy variable for 2021. This enables an assessment of the short-term disruption caused by the pandemic on life expectancy across countries. By integrating structural determinants and external shocks within a unified framework, the study provides a more nuanced understanding of the drivers of population health outcomes.

Overall, this research contributes to the existing literature by offering empirical evidence on the relative importance of economic and social factors in shaping life expectancy. The findings are expected to inform policymakers by highlighting the importance of promoting sustainable economic growth, investing in education, and improving the efficiency of healthcare systems

2. Literature Review

A substantial body of literature has examined the relationship between economic development and population health outcomes, with life expectancy widely recognised as a key indicator of well-being. Early empirical studies established a strong positive association between income levels, typically measured by GDP per capita, and life expectancy across countries. However, subsequent research has demonstrated that this relationship is complex and mediated by a range of socio-economic, institutional, and policy-related factors.

One of the most influential contributions in this field is the Preston Curve, introduced by Preston (1975), which illustrates the non-linear relationship between life expectancy and per capita income. The Preston Curve shows that life expectancy increases rapidly at lower levels of income but exhibits diminishing returns as income rises. This implies that economic growth plays a more significant role in improving health outcomes in low-income countries, while its marginal impact declines in high-income settings. Moreover, shifts in the curve over time indicate that improvements in medical technology, public health interventions, and knowledge diffusion have played a crucial role in enhancing life expectancy beyond income effects alone.

Building on this perspective, Angus Deaton (2004, 2008) argues that health improvements are not solely driven by income growth but also by the spread of medical knowledge, globalisation, and technological progress. His work highlights that substantial gains in life expectancy can occur even in lower-income countries when effective healthcare systems and access to innovations are present. Similarly, David E. Bloom and David Canning (2007) emphasise the bidirectional relationship between health and economic growth, suggesting that improvements in health outcomes can themselves act as a driver of economic development.

In parallel, the role of human capital, particularly education, has received considerable attention in the literature. Gary S. Becker et al. (2005) argue that education enhances individuals' ability to make informed health-related decisions, adopt healthier behaviours, and access better economic opportunities. Empirical evidence further supports a strong and stable positive relationship between education and life expectancy, often more consistent than that of income. provides additional evidence of a bidirectional relationship, indicating that improvements in health can also reinforce educational attainment and human capital accumulation.

Health expenditure has also been widely analysed as a determinant of life expectancy, although the empirical findings remain mixed. David A. Cutler et al. (2006) highlight that increases in healthcare spending can contribute to improved health outcomes, particularly through technological advancements and better treatment methods. However, Christopher J. L. Murray and Julio Frenk (2000) argue that the effectiveness of healthcare systems depends more on efficiency, accessibility, and institutional quality than on expenditure levels alone. This helps explain why higher spending does not always translate into better life expectancy outcomes, particularly in countries with inefficient health systems.

The relationship between unemployment and health outcomes is similarly complex and context-dependent. Higher unemployment is generally associated with reduced income, increased psychological stress, and limited access to healthcare, all of which can negatively affect life expectancy. However, empirical studies, such as those by Christopher J. Ruhm (2000), suggest that the relationship may not always be straightforward, as economic downturns can sometimes be associated with temporary improvements in certain health outcomes due to changes in behaviour and lifestyle patterns. These mixed findings indicate that labour market conditions interact with broader institutional and social factors in shaping health outcomes.

More recent studies have increasingly employed panel data techniques to better capture both cross-country variation and time dynamics. For instance, Owen O'Donnell et al. (2015) and Ellen Van Doorslaer (2014) utilise panel regression frameworks to demonstrate that income and education remain robust determinants of life expectancy, while the effects of healthcare expenditure and unemployment vary across model specifications and institutional contexts. Panel models, particularly fixed effects and two-way fixed effects, allow researchers to control for unobserved heterogeneity across countries and over time, thereby providing more reliable and consistent estimates.

Despite the extensive literature, several important gaps remain. First, many studies focus on a limited set of variables, often analysing income or healthcare expenditure in isolation rather than considering a broader set of socio-economic determinants within a unified framework. Second, relatively few studies explicitly incorporate global shocks, such as the COVID-19 pandemic, when analysing life expectancy trends, despite their significant impact on mortality and healthcare systems worldwide. While the existing literature emphasises the importance of institutional quality and healthcare efficiency, this study focuses on observable macro-level

indicators due to data availability constraints, while still capturing key dimensions of economic and social development.

This study contributes to the literature by integrating economic and social determinants of life expectancy, GDP per capita, health expenditure, education, and unemployment, within a comprehensive panel data framework covering 142 countries over the period 2010-2023. Furthermore, it explicitly accounts for the impact of the COVID-19 pandemic through the inclusion of a 2021 dummy variable. By employing pooled OLS, fixed effects, and two-way fixed effects models, the analysis provides a more nuanced and robust understanding of the determinants of life expectancy, while addressing key gaps in the existing literature.

3. Data

3.1 Independent and Dependent Variable

The objective of this study is to analyse the relationship between economic development and population health outcomes across countries. In particular, the analysis focuses on the impact of income levels on life expectancy, while controlling for key socio-economic factors that may influence health outcomes.

The dependent variable in this study is life expectancy, measured in years, which represents the average number of years a newborn is expected to live under prevailing mortality conditions. As a widely used indicator of population health, life expectancy captures both healthcare effectiveness and broader socioeconomic conditions.

The primary independent variable is GDP per capita, measured in current US dollars, representing the level of economic output per person. Given the expected non-linear relationship between income and life expectancy (Deaton, 2006), this variable is transformed into its logarithmic form in the econometric analysis. This transformation helps to reduce skewness and allows for a more interpretable elasticity-based relationship between income and health outcomes.

In addition to income, the model incorporates several control variables to provide a more comprehensive framework. Health expenditure, measured as a percentage of GDP, captures the level of investment in healthcare systems. Education, proxied by average years of schooling, reflects the level of human capital and its potential influence on health-related

behaviours. Finally, the unemployment rate, measured as the percentage of the labour force without employment, captures labour market conditions and economic instability.

Overall, the inclusion of these variables allows for a multidimensional analysis of the determinants of life expectancy, accounting for both economic and social factors.

3.2 Source of Data and Descriptive Statistics

The dataset used in this study is obtained from internationally recognised sources, primarily the World Bank Open Data. The final dataset consists of an unbalanced panel of 142 countries observed over the period 2010-2023, resulting in a total of 1,649 observations.

The panel structure combines both cross-sectional and time-series dimensions, enabling the analysis to capture variations across countries as well as changes over time. This is particularly important in the context of life expectancy, which evolves gradually and is influenced by both structural and temporal factors.

Summary statistics for the main variables are presented in Table 1.

Table 1: Summary Statistics of Variables

Variable	Mean	Std. Dev.	Minimum	Maximum
Life Expectancy	73.56	7.29	46.05	84.56
GDP per capita	27940.84	25585.37	991.04	145590.76
Health Expenditure	6.76	2.61	1.60	18.52
Education	89.13	25.76	12.81	164.08
Unemployment	7.55	5.54	0.10	36.47

Summary statistics for the main variables are presented in Table 1. Life expectancy has an average value of approximately 73.6 years, with a relatively wide range from 46.05 to 84.56 years. This variation reflects substantial disparities in health outcomes across countries, likely driven by differences in income levels, healthcare systems, and socio-economic conditions.

GDP per capita exhibits significant dispersion, with values ranging from less than 1,000 USD to over 145,000 USD. This high level of inequality across countries provides a strong basis for examining the relationship between economic development and life expectancy. The large standard deviation further indicates the presence of extreme values, justifying the use of a logarithmic transformation in the econometric analysis.

Health expenditure shows moderate variation, with an average of approximately 6.76% of GDP. This suggests differences in healthcare investment across countries, although the relatively narrower range compared to GDP indicates less extreme disparities. Education levels and unemployment rates also display considerable variation, reflecting differences in human capital development and labour market conditions.

Overall, the observed variation in the dataset supports the use of panel data methods, as it allows for a richer analysis of both cross-country differences and temporal dynamics in life expectancy.

3.3 Visual Representation and Distribution of Variables

To further examine the properties of the data, histograms with density curves are presented for the key variables.

The distribution of GDP per capita is highly right-skewed, indicating that a large number of countries have relatively low to moderate income levels, while a small number of countries exhibit extremely high income. This skewness highlights the presence of global income inequality and reinforces the need for a logarithmic transformation in the regression analysis to obtain a more symmetric distribution for subsequent econometric analysis. This transformation is directly incorporated into the regression model, allowing the estimated relationship between income and life expectancy to capture diminishing marginal effects, consistent with the theoretical predictions of the Preston Curve (Deaton, 2006).

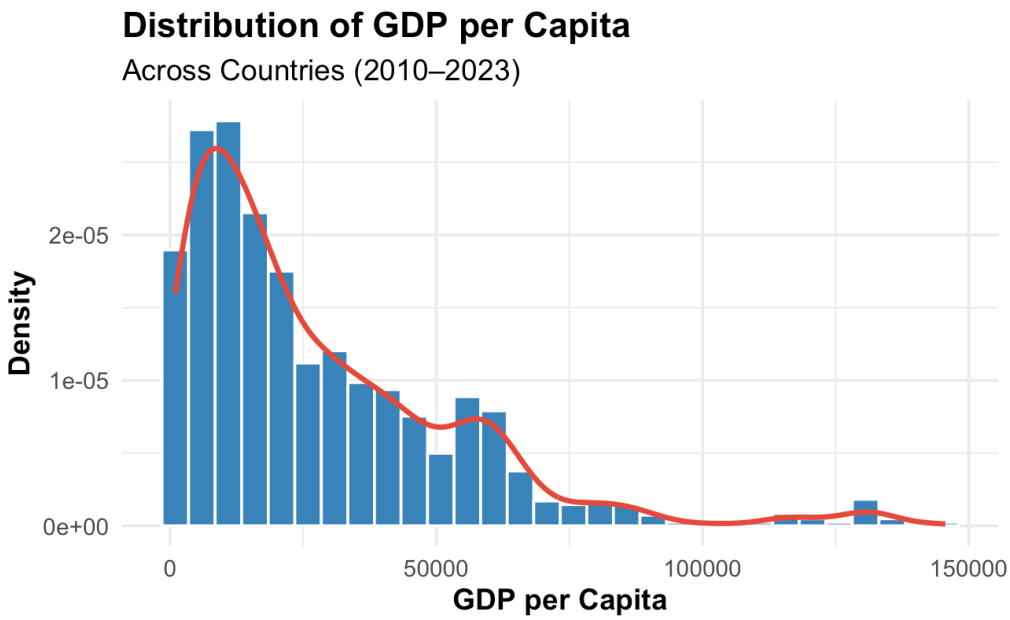


FIGURE 1: Distribution of GDP per Capita

Health expenditure also displays a moderate degree of skewness, reflecting variation in healthcare investment across countries. However, compared to GDP per capita, the distribution is less extreme, suggesting that healthcare spending is relatively more evenly distributed.

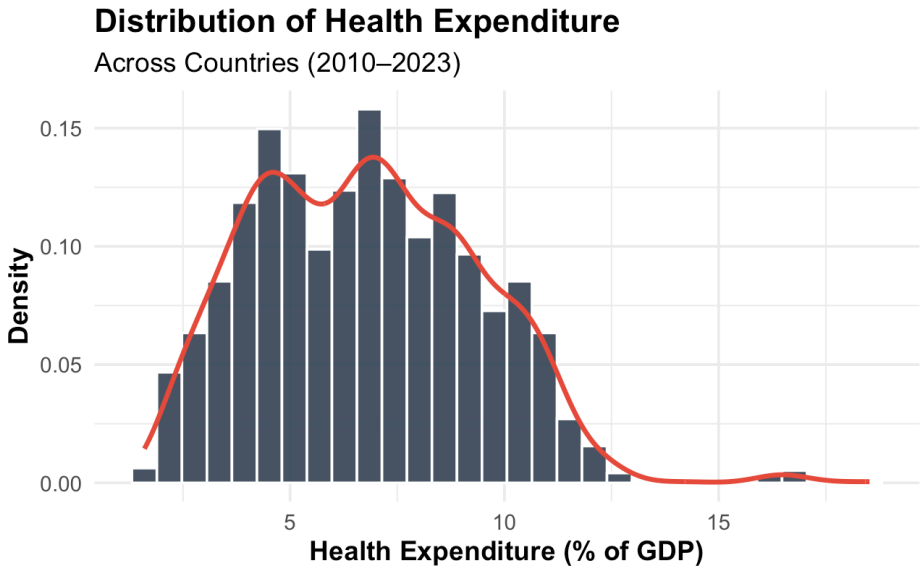


FIGURE 2: Distribution of Health Expenditure

The distribution of education is concentrated around higher values, indicating that many countries have achieved moderate to high levels of schooling. This clustering suggests that variations in education may capture more subtle differences in human capital rather than extreme disparities.

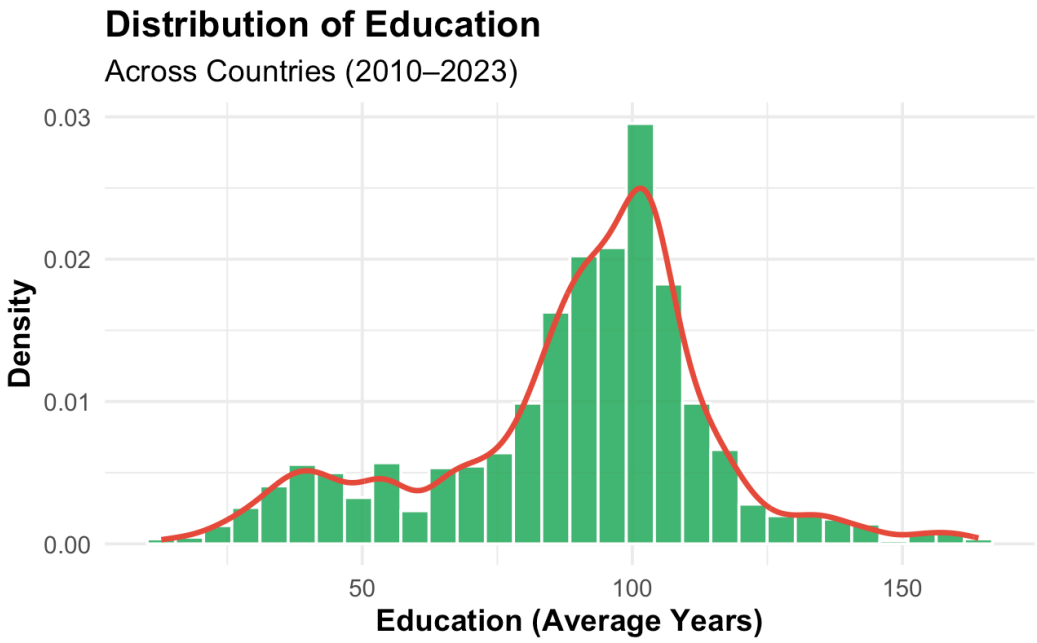


FIGURE 3: Distribution of Education

Life expectancy exhibits a slightly left-skewed distribution, with most countries clustered at relatively high levels. This reflects overall global improvements in health outcomes over time, although a small number of countries still experience significantly lower life expectancy.

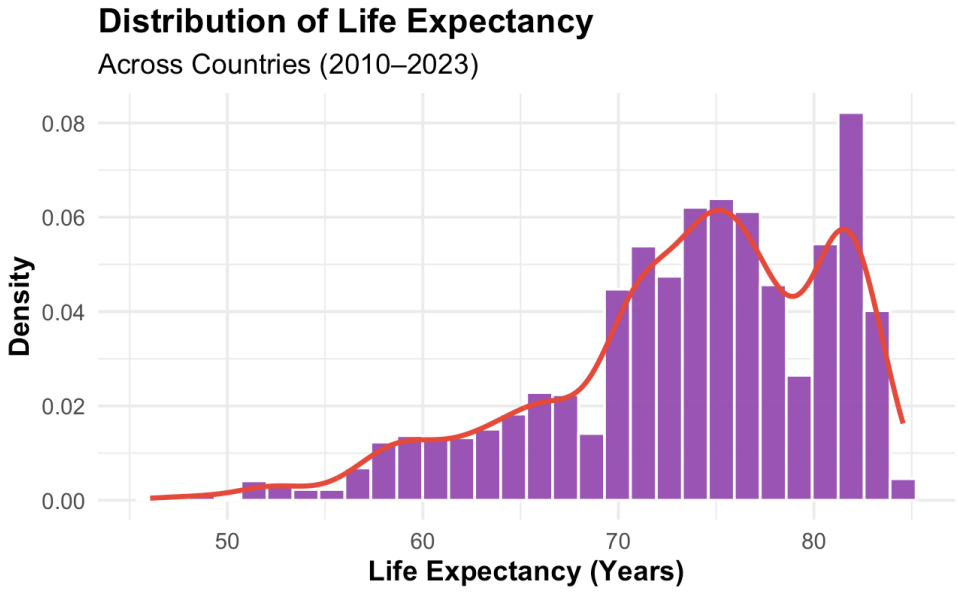


FIGURE 4: Distribution of Life Expectancy

These distributional characteristics are important for the econometric analysis, as they justify variable transformations and provide insights into the underlying structure of the data.

3.4 Correlation Analysis

To examine the relationships between variables, a correlation matrix is computed.

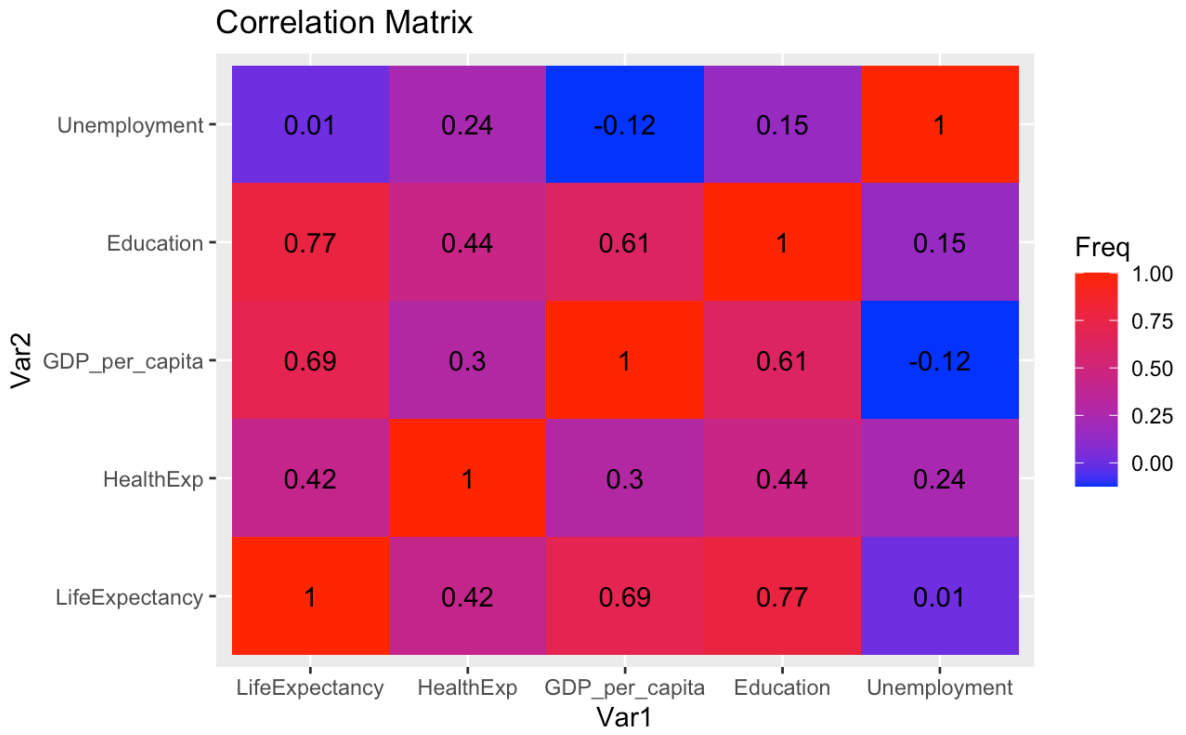


Figure 5: Correlation Matrix

The results indicate a strong positive correlation between GDP per capita and life expectancy, supporting the hypothesis that higher income levels are associated with better health outcomes. Education also exhibits a strong positive relationship with life expectancy, highlighting the role of human capital in improving population health. Health expenditure shows a moderate positive correlation, suggesting that increased healthcare spending is generally associated with improved life expectancy, although the relationship is less pronounced.

In contrast, unemployment displays a relatively weak correlation with life expectancy, indicating that its direct effect may be limited in the descriptive analysis. However, this does not preclude more complex relationships that may emerge in the regression framework.

Additionally, the presence of relatively high correlations among some independent variables, particularly between GDP per capita and education, suggests the potential for multicollinearity. This issue is addressed in the econometric analysis through the use of fixed effects models, which help to mitigate bias arising from omitted variables and correlated regressors.

3.5 Additional Insights from Graphical Analysis

The scatter plot between GDP per capita and life expectancy reveals a clear positive relationship, with higher income levels associated with longer life expectancy. However, the relationship appears non-linear, with diminishing marginal effects at higher income levels. This observation is consistent with the Preston Curve and further supports the use of a logarithmic transformation of GDP per capita.

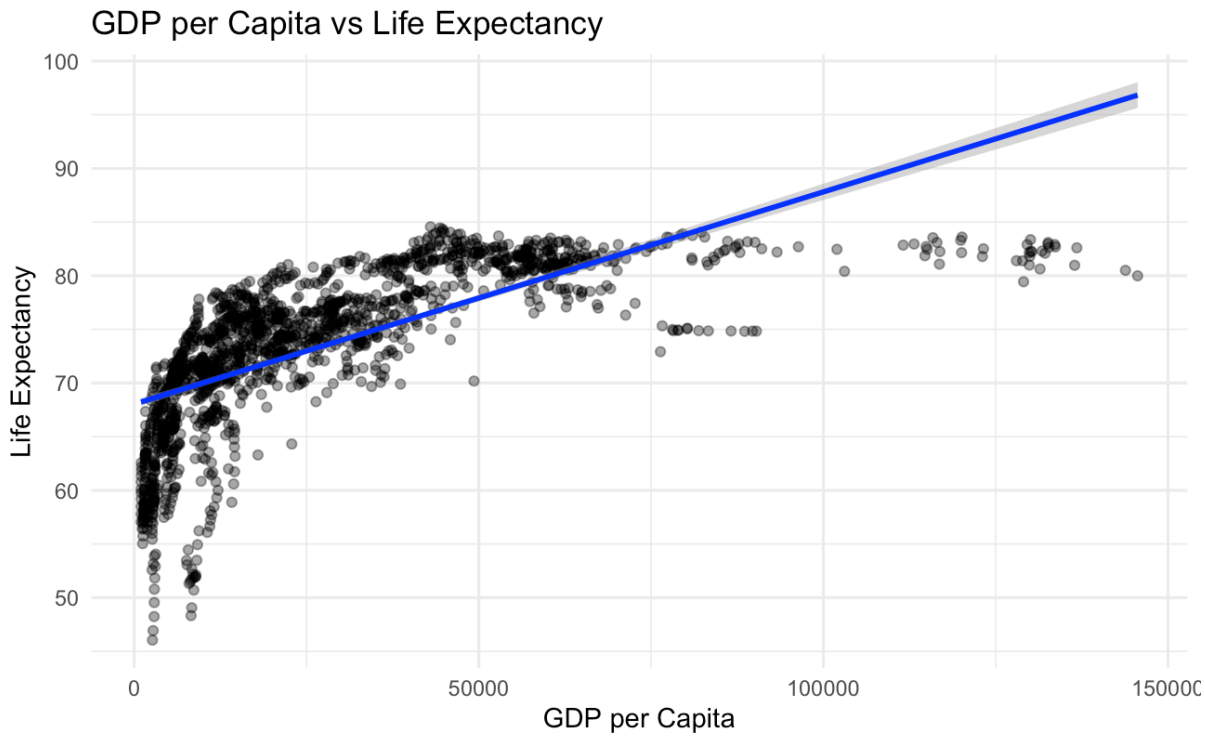


FIGURE 6: GDP per Capita Vs Life Expectancy

The time-series analysis of average life expectancy over the period 2010-2023 shows a general upward trend, reflecting global improvements in healthcare, living standards, and technological progress. However, a noticeable decline is observed during 2020 and 2021, corresponding to the COVID-19 pandemic.

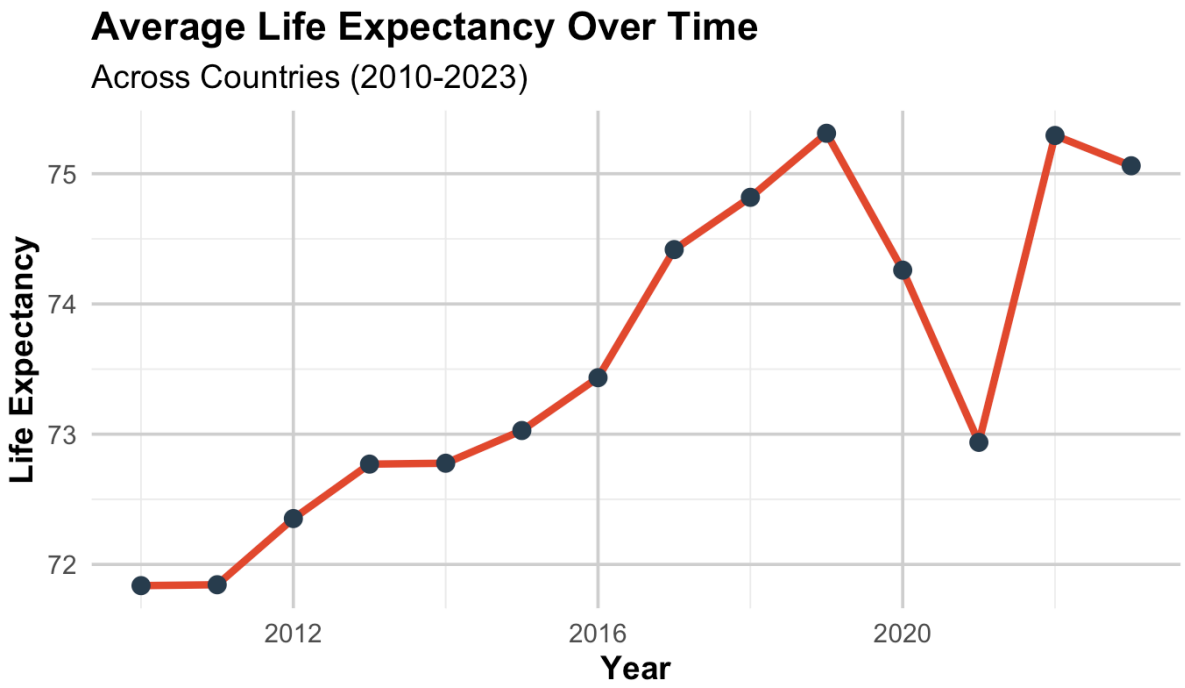


FIGURE 7: Average Life Expectancy Over Time

This pattern highlights the importance of accounting for global shocks in the analysis. The temporary decline followed by partial recovery suggests that while long-term trends in life expectancy are positive, they remain vulnerable to external disruptions.

Overall, the graphical analysis reinforces the expected relationships between economic and social determinants and life expectancy, while also providing preliminary evidence of non-linearity and temporal shocks that are further explored in the econometric models.

4. Model

4.1 Model Framework

To examine the relationship between economic and social determinants and life expectancy, this study employs panel data regression techniques. Panel data provides a significant advantage over purely cross-sectional or time-series data, as it allows for the control of unobserved heterogeneity across countries while also capturing temporal dynamics.

The dataset consists of an unbalanced panel of 142 countries observed over the period 2010-2023. This structure enables the analysis to account for both cross-country differences and changes over time in life expectancy. Formally, this panel structure allows us to control for unobserved country-specific heterogeneity and common time effects, which is crucial for obtaining consistent estimates of the relationship between economic and social determinants and life expectancy.

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4.2 Model Specification

Following the theoretical framework, the baseline econometric model is specified as:
(1)

$$LifeExpectancy_{it} = \beta_0 + \beta_1 log(GDP_{it}) + \beta_2 HealthExp_{it} + \beta_3 Education_{it} + \beta_4 Unemployment_{it} + \varepsilon_{it}$$

where $i = 1, \ldots, N$ indexes countries and $t = 1, \ldots, T$ indexes time. The dependent variable $LifeExpectancy_{it}$ represents average life expectancy. The key explanatory variable is the logarithm of GDP per capita, $\log(GDP_{it})$, which captures income effects, while $HealthExp_{it}$, $Education_{it}$, and $Unemployment_{it}$ are included as control variables. The error term ε_{it} captures unobserved factors affecting life expectancy.

The logarithmic transformation of GDP per capita is applied to account for potential non-linearities and to allow for an elasticity-based interpretation of the relationship between income and life expectancy. The coefficients β_1, β_2, β_3, β_4 measure the marginal effects of the respective variables on life expectancy. This specification allows us to estimate the partial effect of each explanatory variable on life expectancy, holding other factors constant.

Equation (1) represents the baseline specification, which serves as the foundation for the fixed effects and two-way fixed effects specifications estimated in the subsequent analysis.

4.3 Panel Models Estimated

To ensure the robustness of the empirical analysis, three alternative panel data models are estimated: pooled ordinary least squares (OLS), fixed effects (FE), and two-way fixed effects models. Each specification addresses different econometric concerns and allows for a more comprehensive understanding of the relationship between the variables.

The pooled OLS model serves as a baseline specification, assuming a common intercept across all countries and ignoring unobserved heterogeneity. While this approach is straightforward, it may produce biased and inconsistent estimates if country-specific characteristics are correlated with the explanatory variables.

To address this limitation, a fixed effects model is employed. This specification introduces country-specific intercepts, thereby controlling for time-invariant unobserved heterogeneity, such as institutional quality, geography, and cultural factors. By focusing on within-country variation over time, the fixed effects model provides more reliable estimates.

To further improve the specification, a two-way fixed effects model is estimated by incorporating both country-specific and time-specific effects. This approach allows the model to account for global shocks and common trends affecting all countries simultaneously, such as technological progress, economic crises, and pandemics. As a result, the two-way fixed effects model provides the most comprehensive control for unobserved heterogeneity.

These alternative specifications allow us to assess how controlling for unobserved heterogeneity affects both the magnitude and interpretation of the estimated coefficients.

4.4 Fixed Effects and Two-Way Fixed Effects Specifications

Extending equation (1), the fixed effects model is specified as:

(2)

$$LifeExpectancy_{it} = \beta_0 + \beta_1 log(GDP_{it}) + \beta_2 HealthExp_{it} + \beta_3 Education_{it} + \beta_4 Unemployment_{it} + \alpha_i + \varepsilon_{it}$$

where α_i represents country-specific effects that capture time-invariant characteristics.

The two-way fixed effects model extends equation (2) by including time-specific effects:

(3)

$$LifeExpectancy_{it} = \beta_0 + \beta_1\, log(GDP_{it}) + \beta_2\, HealthExp_{it} + \beta_3\, Education_{it} + \beta_4\, Unemployment_{it} + \alpha_i + \lambda_t + \varepsilon_{it}$$

where λ_t captures time-specific effects common to all countries.

These specifications progressively control for unobserved heterogeneity, allowing for a more robust identification of the relationship between the explanatory variables and life expectancy by controlling for both time-invariant country characteristics and common time shocks.

4.5 COVID-19 Shock Specification

To capture the impact of the COVID-19 pandemic on life expectancy, a dummy variable for the year 2021 is introduced. This variable takes the value of 1 for the year 2021 and 0 otherwise, allowing the model to isolate the short-term effects of the pandemic.

To capture short-term exogenous shocks that are not fully absorbed by country fixed effects, equation (2) is augmented with a COVID-19 dummy variable:

(4)

$$LifeExpectancy_{it} = \beta_0 + \beta 1 log(GDP_{it}) + \beta_2 HealthExp_{it} + \beta_3 Education_{it} + \beta_4 Unemployment_{it} + \delta\ COVID2021_{it} + \alpha_i + \varepsilon_{it}$$

Since the model does not include time fixed effects, the COVID-19 dummy captures a specific time shock affecting all countries in 2021. The dummy is included only in the fixed effects specification and not in the two-way fixed effects model, as time fixed effects would fully absorb year-specific variation, leading to perfect multicollinearity.

4.6 Addressing Endogeneity (Lagged GDP)

A potential concern in estimating the relationship between income and life expectancy is the presence of endogeneity, particularly due to reverse causality and simultaneity bias. While higher income may improve health outcomes, better health can also contribute to economic growth.

To address this issue, a lagged value of GDP per capita is introduced into the model. By using past income levels as an explanatory variable, this approach helps mitigate reverse causality and provides a more reliable estimate of the impact of economic development on life expectancy.

The fixed effects model is re-estimated using lagged GDP per capita, thereby reducing potential simultaneity bias and allowing for a more robust interpretation of the relationship between income and life expectancy.

4.7 Model Selection and Justification

The choice of model specification is guided by the objective of obtaining consistent and unbiased estimates. While the pooled OLS model provides a useful benchmark, it fails to

account for unobserved heterogeneity across countries, which may lead to biased estimates if such factors are correlated with the explanatory variables.

The fixed effects model addresses this issue by controlling for time-invariant country-specific characteristics. Unlike pooled OLS, which primarily exploits cross-country variation, the fixed effects approach relies on within-country variation over time, thereby improving the reliability of the estimated relationships.

Building on this framework, the two-way fixed effects model further enhances the specification by accounting for both country-specific and time-specific effects. The inclusion of time fixed effects removes common shocks affecting all countries simultaneously, such as global economic trends or the COVID-19 pandemic.

As a result, the two-way fixed effects model improves the credibility of the estimates by reducing potential omitted variable bias arising from both unobserved country characteristics and time-specific factors. Consequently, the coefficients can be interpreted as reflecting within-country variation.

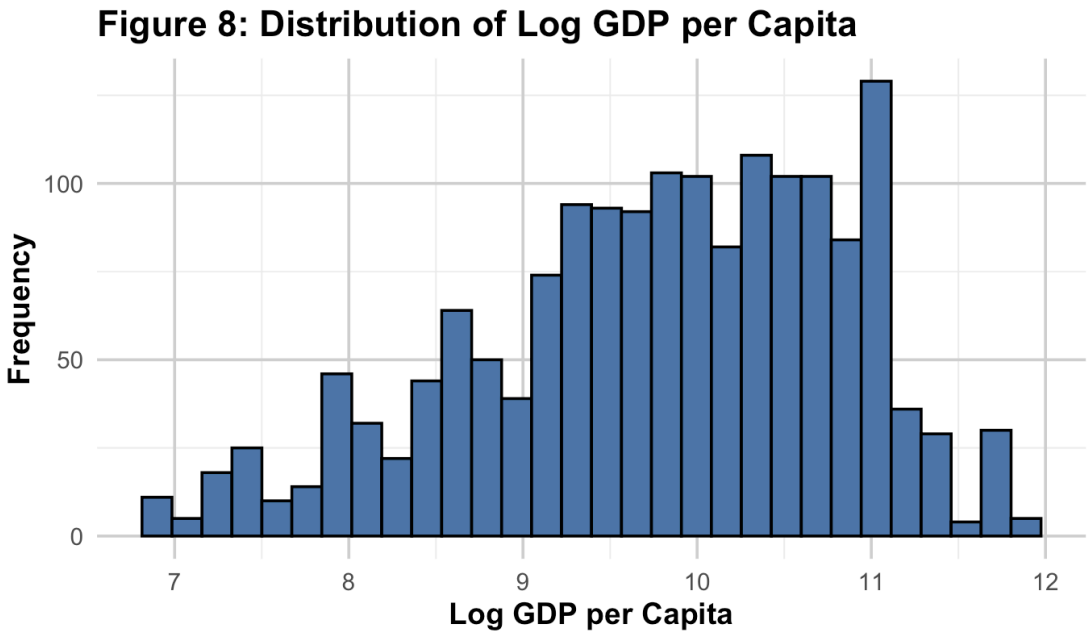
Therefore, the two-way fixed effects model is considered the preferred specification in this study, as it provides the most appropriate framework for analysing the determinants of life expectancy while accounting for both unobserved heterogeneity and common temporal shocks.

5. Result

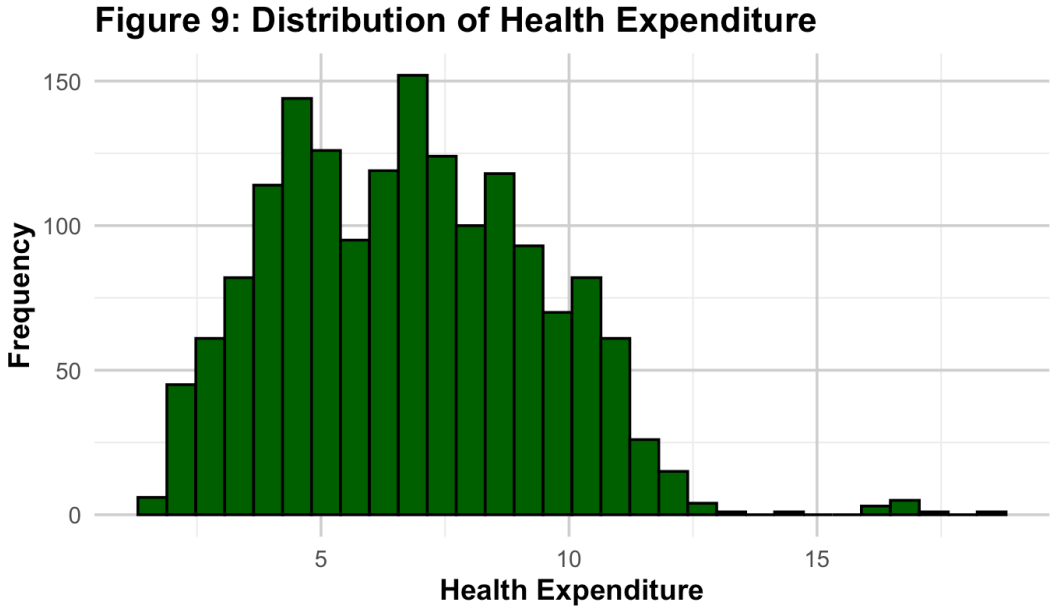
5.1 Descriptive and Distribution Analysis

To provide an initial understanding of the data, the distribution of key variables is examined using graphical analysis.

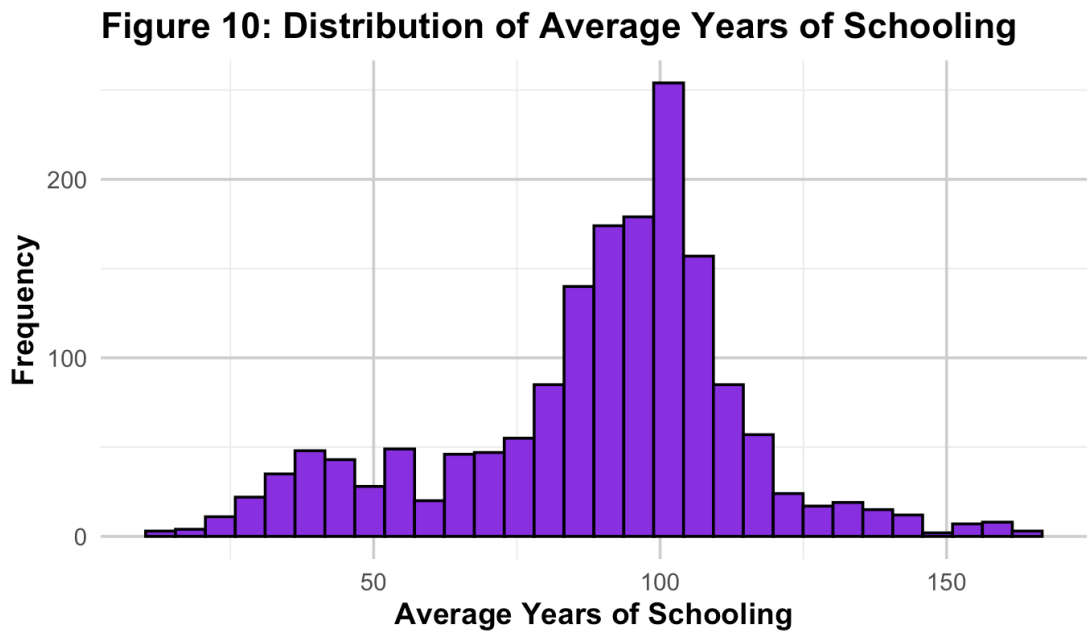
The distribution of log GDP per capita appears approximately normal, indicating that the logarithmic transformation successfully reduces the strong right-skewness observed in the raw GDP variable. This transformation is important for the econometric analysis, as it ensures a more symmetric distribution and improves the reliability of regression estimates.



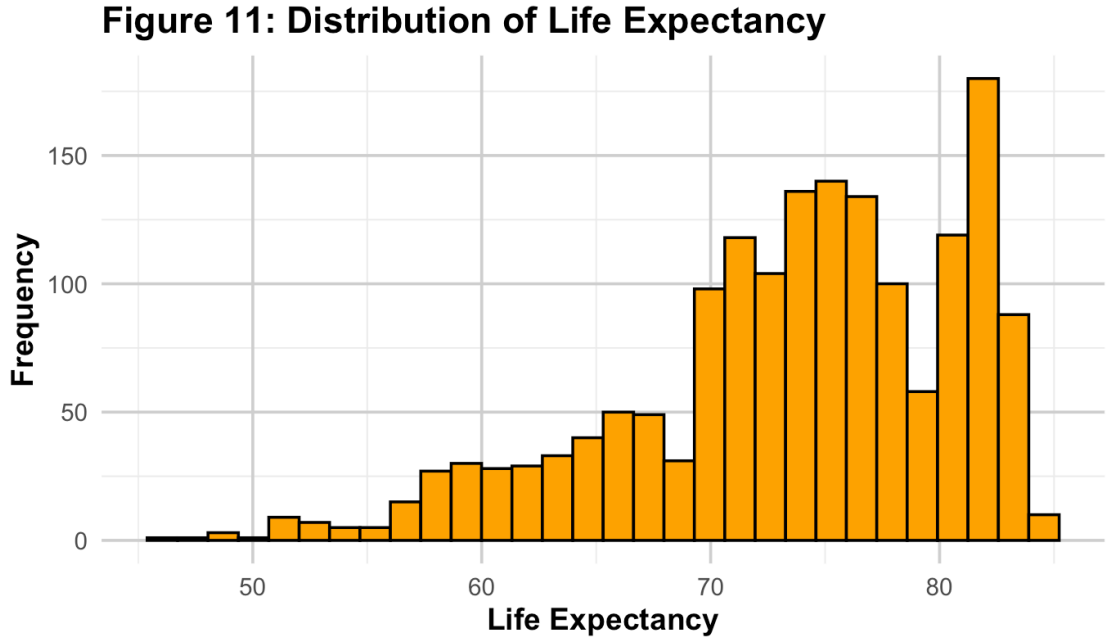
Health expenditure exhibits a slightly right-skewed distribution, suggesting that while most countries allocate moderate shares of GDP to healthcare, a smaller number of countries spend disproportionately higher amounts. This variation reflects differences in healthcare systems and fiscal priorities across countries.



The distribution of education, measured by average years of schooling, shows considerable variation across countries, with a concentration around medium to higher levels. This indicates that while many countries have achieved substantial improvements in education, disparities in human capital development persist.



Life expectancy displays a relatively concentrated distribution, with most observations falling within the range of 65 to 80 years . This suggests that global health outcomes have improved over time, although variation across countries remains evident.



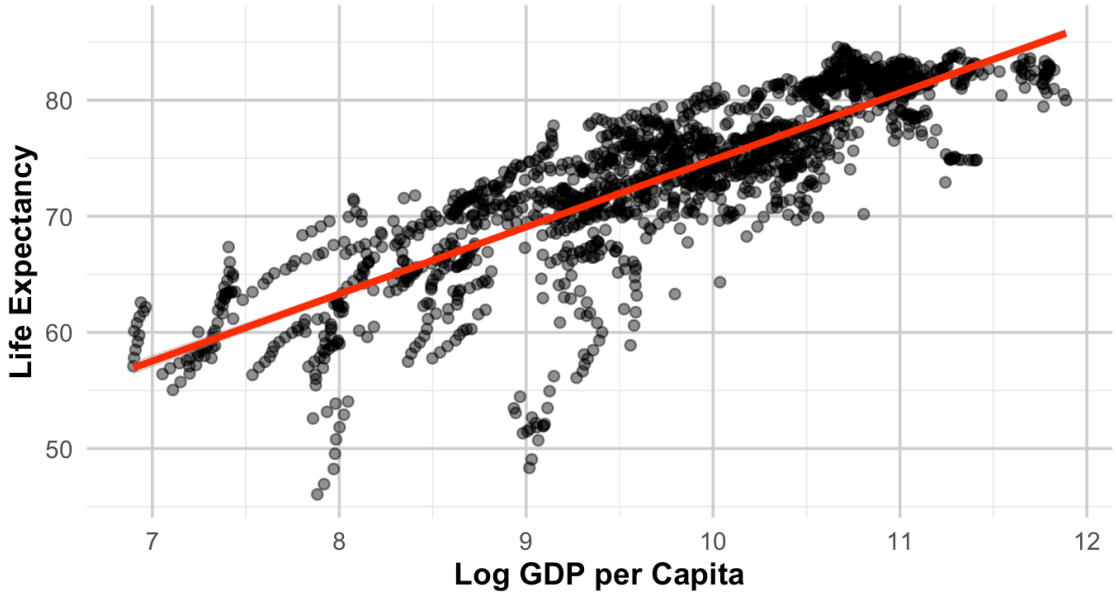
Overall, the distributional analysis confirms the presence of variation across countries and supports the use of panel data techniques for further analysis.

5.2 Bivariate Relationships

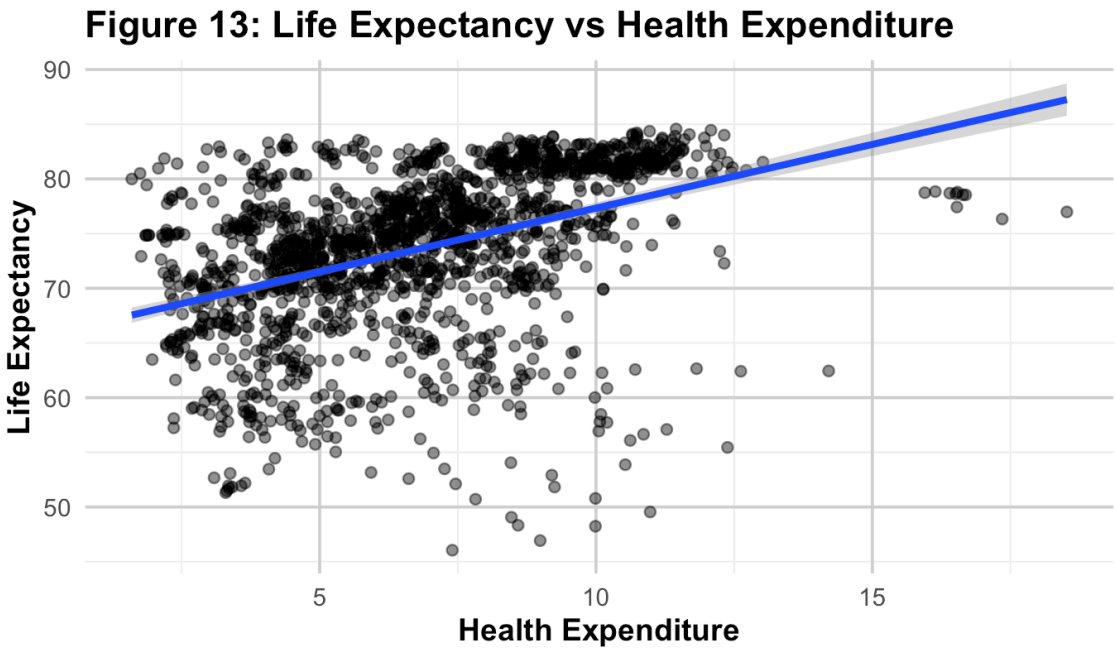
To explore the relationships between the key variables, scatter plots are analysed.

The relationship between life expectancy and log GDP per capita reveals a clear positive association, indicating that higher income levels are generally linked to improved health outcomes. However, the relationship appears non-linear, with diminishing marginal effects at higher income levels. This pattern is consistent with the Preston Curve (Deaton, 2006) and supports the use of a logarithmic transformation in the regression model.

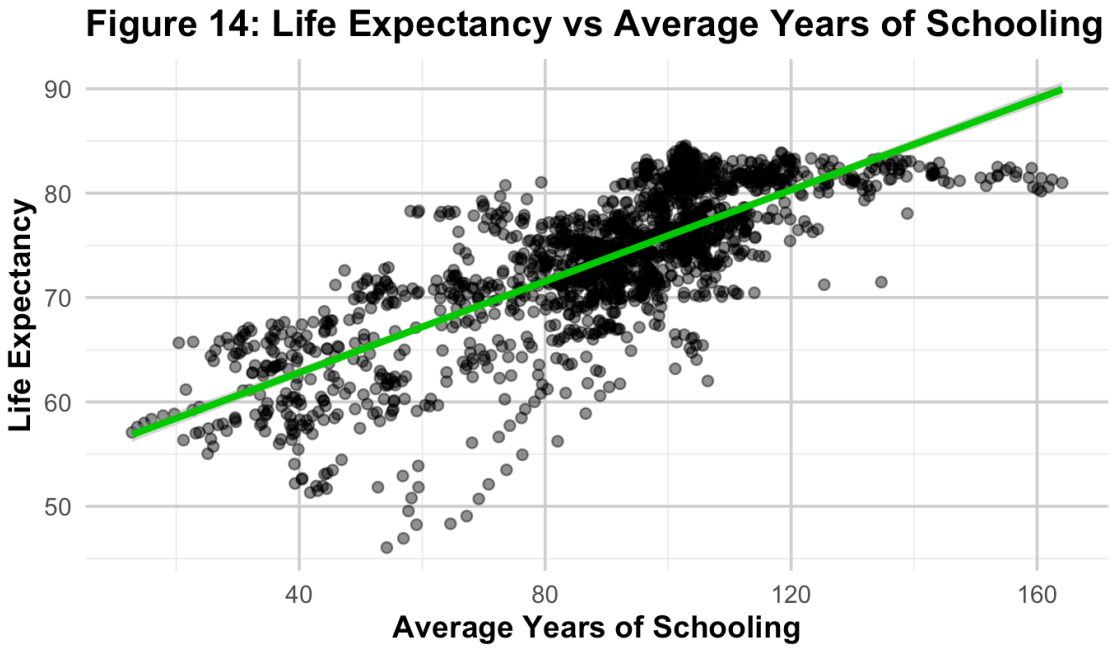
Figure 12: Life Expectancy vs Log GDP per Capita



A positive relationship is also observed between health expenditure and life expectancy, suggesting that higher investment in healthcare is associated with better health outcomes. Nevertheless, the dispersion of data points indicates that the effectiveness of health expenditure varies across countries, pointing towards differences in efficiency rather than spending levels alone.



Similarly, education shows a strong and consistent positive relationship with life expectancy. Countries with higher levels of schooling tend to exhibit higher life expectancy, highlighting the importance of human capital in shaping health outcomes.

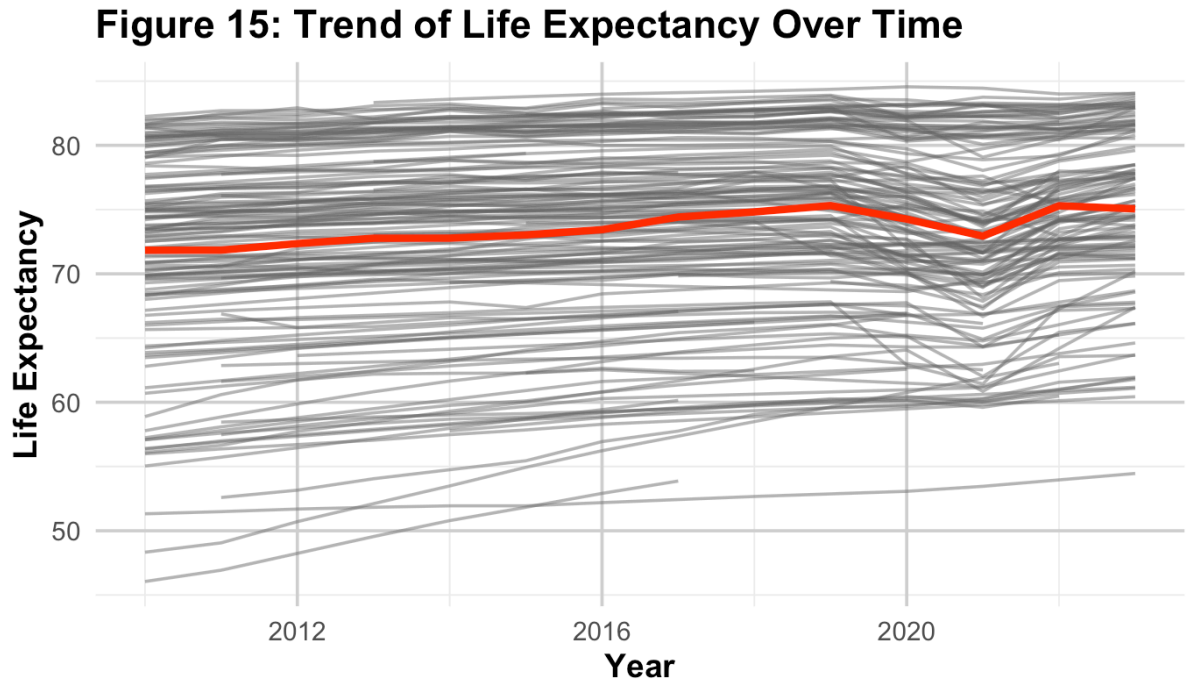


These bivariate relationships provide preliminary evidence supporting the expected theoretical links, although they do not account for confounding factors, which are addressed in the regression analysis.

5.3 Time Trend Analysis

To examine the evolution of life expectancy over time, a time-series analysis is conducted.

The results show a steady upward trend in global life expectancy from 2010 to 2019, reflecting improvements in healthcare systems, living standards, and technological advancements. However, a noticeable decline is observed during 2020 and 2021, corresponding to the COVID-19 pandemic. This pattern is further confirmed by the panel regression analysis, where the COVID-19 dummy variable for 2021 shows a statistically significant negative effect on life expectancy.



Following this decline, a partial recovery is observed in subsequent years, indicating that the effects of the pandemic, while significant, may be temporary. This pattern highlights the importance of accounting for global shocks in the econometric analysis.

5.4 Panel Regression Results

To formally assess the determinants of life expectancy, panel regression models are estimated using pooled OLS, fixed effects, and two-way fixed effects specifications. The results are presented in Table 2.

Variables	Pooled OLS	Fixed Effects	Two-Way FE
Log GDP	4.121***	4.045***	0.684*
	(0.144)	(0.294)	(0.290)
Health Expenditure	0.297***	-0.109**	-0.150***
	(0.039)	(0.036)	(0.033)
Education	0.073***	0.044***	0.022***
	(0.006)	(0.005)	(0.004)
Unemployment	-0.106***	0.045**	0.080***
	(0.017)	(0.015)	(0.012)
Observations	1649	1649	1649
R^2	0.748	0.210	0.071
Adjusted R^2	0.747	0.133	-0.027
Fixed Effects	No	Yes	Yes
Time Effects	No	No	Yes

NOTE: *p < 0.1; **p < 0.05; ***p < 0.01

The pooled OLS results indicate that all explanatory variables are statistically significant. GDP per capita exhibits a strong positive effect on life expectancy, suggesting that higher income levels are associated with improved health outcomes. This finding is consistent with the Preston Curve hypothesis and aligns with existing empirical studies, which highlight the strong but diminishing impact of income on life expectancy (Lynch & Kaplan, 1997). Health expenditure and education also show positive and significant effects, while unemployment has a negative impact.

However, the pooled OLS model does not account for unobserved heterogeneity across countries, which may bias the estimates.

The differences across pooled OLS, fixed effects, and two-way fixed effects models highlight the importance of controlling for unobserved heterogeneity in panel data analysis. While pooled OLS primarily captures cross-country variation, the fixed effects model relies on within-country variation over time, which can lead to changes in coefficient magnitude and

interpretation. The two-way fixed effects model further accounts for time-specific shocks affecting all countries simultaneously. As a result, the differences in estimated coefficients across models suggest that the observed relationships should be interpreted with caution in terms of causality, as they may reflect both structural differences across countries and dynamic changes within countries over time.

The fixed effects model provides a more robust specification by controlling for time-invariant country-specific factors. The results show that GDP per capita and education remain positively significant, confirming their strong influence on life expectancy. In contrast, health expenditure becomes negative and statistically significant, suggesting that higher spending does not necessarily translate into better outcomes once country-specific characteristics are controlled for. This result highlights an important distinction between the level and the quality of healthcare expenditure. While higher spending reflects greater financial resources allocated to healthcare, it does not guarantee effective service delivery or improved outcomes. Inefficiencies in resource allocation, institutional constraints, and differences in healthcare system performance may lead to situations where increased expenditure fails to improve health outcomes. Therefore, this finding highlights the importance of efficiency in healthcare systems and suggests that healthcare outcomes depend more on how resources are utilised rather than the absolute level of expenditure.

The coefficient on unemployment becomes positive in the fixed effects model, indicating that within-country variations may capture different socio-economic dynamics compared to cross-country comparisons. This result suggests that the relationship between unemployment and health outcomes is complex and context-dependent.

The two-way fixed effects model further incorporates time-specific effects, capturing global shocks and common trends. The magnitude of the GDP coefficient decreases, indicating that part of its effect is absorbed by time-specific factors. Education remains positive and statistically significant, reinforcing its role as a key determinant of life expectancy. Health expenditure continues to exhibit a negative and significant effect, supporting the argument that efficiency matters more than spending levels. Unemployment also remains positive and significant, reflecting broader structural dynamics.

Overall, the comparison across models highlights the importance of controlling for both country-specific and time-specific heterogeneity in panel data analysis.

5.5 Impact of COVID-19 *(Dummy Variable Analysis)*

To examine the impact of the COVID-19 pandemic, a dummy variable for the year 2021 is included in the fixed effects model. The results are presented in Table 3.

Variables	Model (COVID 2021)
Log GDP	4.303***
	(0.289)
Health Expenditure	0.004
	(0.038)
Education	0.046***
	(0.004)
Unemployment	0.052***
	(0.014)
COVID Dummy	-0.646***
	(0.080)
Observations	1649
R^2	0.243

Note: *p < 0.1; **p < 0.05; ***p < 0.01

The coefficient on the COVID-19 dummy variable is negative and statistically significant, indicating a substantial decline in life expectancy during 2021. This finding reflects the widespread impact of the pandemic on mortality rates, healthcare systems, and overall population health.

The significance of the COVID-19 variable suggests that the effects of the pandemic were particularly pronounced in 2021, consistent with the timing of major infection waves and

healthcare system pressures across countries. This result highlights the importance of accounting for global shocks when analysing health outcomes.

5.6 Interpretation of Results

The empirical findings provide strong evidence that both economic and social factors play a significant role in determining life expectancy.

GDP per capita emerges as a robust and consistent determinant across all model specifications, confirming that higher income levels contribute to improved health outcomes. Education also shows a strong and stable positive effect, highlighting the importance of human capital in shaping health behaviours and access to healthcare.

In contrast, health expenditure exhibits mixed effects, becoming negative in models that control for unobserved heterogeneity. This suggests that the effectiveness of healthcare systems depends not only on the level of spending but also on how resources are allocated and utilised.

The relationship between unemployment and life expectancy appears complex, with differing effects across model specifications. This indicates that labour market conditions interact with broader socio-economic factors in influencing health outcomes.

Finally, the analysis of the COVID-19 dummy variable confirms that global shocks can have significant short-term effects on life expectancy, with a pronounced negative impact observed in 2021.

Overall, the results highlight the importance of adopting a comprehensive approach to analysing health outcomes, taking into account both economic development and social determinants.

5.7 Robustness Check: Lagged GDP

To further address potential endogeneity concerns arising from reverse causality, the fixed effects model is re-estimated using a lagged value of GDP per capita. The results show that lagged GDP remains positive and highly statistically significant, confirming that past income levels have a strong and persistent effect on life expectancy.

The consistency of this result strengthens the robustness of the baseline findings, suggesting that the positive relationship between economic development and life expectancy is not driven by reverse causality. Overall, the inclusion of lagged GDP reinforces the validity of the empirical results and provides stronger evidence consistent with a causal interpretation of the income–health relationship.

5.8 Heterogeneity Analysis

To examine whether the determinants of life expectancy differ across countries at different levels of development, a heterogeneity analysis is conducted by dividing the sample into high-income and low- and middle-income countries to account for structural heterogeneity across countries.

The fixed effects model is estimated separately for each subgroup. The results reveal important differences across the two groups. GDP per capita remains positive and highly significant in both high-income and low-income countries, indicating that economic growth consistently improves life expectancy.

However, the role of other variables differs substantially. Education has a stronger and more statistically significant effect in low-income countries, suggesting that improvements in human capital play a more crucial role in enhancing health outcomes in developing economies. In high-income countries, the effect of education remains positive but is comparatively smaller.

Health expenditure shows weaker and less consistent effects in high-income countries, possibly reflecting diminishing returns in well-developed healthcare systems. In contrast, in low-income countries, health expenditure is statistically significant but exhibits a negative coefficient, which may reflect inefficiencies in healthcare systems, reverse causality, or reactive increases in health spending in response to poor health outcomes.

Unemployment does not appear to have a statistically significant effect in high-income countries but becomes significant in low-income countries, indicating that labour market conditions are more critical in shaping health outcomes in less developed economies.

Overall, these findings highlight substantial heterogeneity in the determinants of life expectancy and underscore the importance of designing policy interventions that are tailored to different stages of economic development.

Table 4: Fixed Effects Estimates by Income Group

	High Income FE	Low Income FE
logGDP	3.781	3.985
	(0.657)	(0.322)
HealthExp	0.079	−0.134
	(0.046)	(0.042)
Education	0.010	0.055
	(0.004)	(0.006)
Unemployment	−0.012	0.045
	(0.025)	(0.017)
Num.Obs.	256	1393
R2	0.293	0.220
R2 Adj.	0.222	0.143
AIC	316.2	4162.7
BIC	333.9	4188.9

	High Income FE	Low Income FE
RMSE	0.44	1.07

6. Conclusion

This study examined the relationship between economic and social determinants and life expectancy using a panel dataset of 142 countries over the period 2010-2023. By employing panel data techniques, including pooled OLS, fixed effects, and two-way fixed effects models, the analysis provides a comprehensive assessment of how economic development and socio-economic factors influence population health outcomes across countries. By empirically validating key theoretical predictions from the existing literature within a unified panel data framework, this study strengthens the evidence on the multidimensional determinants of life expectancy.

The empirical findings reveal that GDP per capita and education are robust and consistent determinants of life expectancy. Higher income levels are associated with improved access to healthcare, better living conditions, and enhanced nutrition, all of which contribute to increased longevity. Similarly, education emerges as a key factor in shaping health outcomes, as it promotes greater health awareness, improved decision-making, and more efficient utilisation of healthcare services. The consistency of these results across different model specifications highlights the central role of economic development and human capital in determining life expectancy. Importantly, when potential endogeneity is addressed by introducing lagged GDP per capita, the positive and statistically significant effect of income remains robust, providing stronger evidence that the observed relationship is not driven by reverse causality.

In contrast, the effect of health expenditure appears more nuanced. While a positive relationship is observed in the pooled OLS model, the coefficient becomes negative in fixed effects specifications, suggesting that higher spending does not necessarily translate into better health outcomes once country-specific characteristics are taken into account. This finding emphasises the importance of efficiency in healthcare systems, indicating that the allocation and management of resources may be more important than the overall level of expenditure.

The relationship between unemployment and life expectancy is found to be complex and sensitive to model specification. While unemployment exhibits a negative effect in cross-sectional analysis, it becomes positive in fixed effects models, suggesting that within-country dynamics may differ from cross-country comparisons. This result highlights the importance of considering broader socio-economic and institutional factors when analysing labour market effects on health outcomes.

The analysis also provides evidence on the impact of global shocks. The inclusion of a COVID-19 dummy variable for 2021 reveals a statistically significant negative effect on life expectancy, reflecting the substantial disruption caused by the pandemic. This finding underscores the vulnerability of health outcomes to external shocks and highlights the importance of incorporating such factors into empirical analysis.

From a policy perspective, the results suggest that improving life expectancy requires a multidimensional approach. While economic growth remains important, policymakers should also prioritise investments in education and focus on improving the efficiency of healthcare systems. Increasing healthcare expenditure alone may not be sufficient unless resources are effectively allocated. Additionally, the findings highlight the need for resilient healthcare systems capable of responding to global shocks such as pandemics.

Despite its contributions, this study has several limitations. The model may omit other relevant determinants of life expectancy, such as environmental factors, income inequality, and institutional quality. Furthermore, potential non-linearities and interaction effects are not fully explored. Future research could extend the analysis by incorporating additional variables, exploring alternative model specifications, and examining heterogeneous effects across different country groups.

The heterogeneity analysis further reveals that the impact of key determinants varies across country groups. In particular, education and labour market conditions play a more significant role in low-income countries, while the effects are more moderate in high-income economies. These findings emphasise the importance of designing policies that are tailored to different levels of economic development.

In conclusion, this study provides strong empirical evidence that both economic and social factors play a crucial role in shaping life expectancy. By integrating multiple determinants

within a panel data framework and accounting for global shocks, the analysis contributes to a more comprehensive understanding of population health outcomes and offers valuable insights for both policymakers and researchers.

7. Limitations

A potential limitation of this analysis is the presence of endogeneity, particularly between income and life expectancy, as improvements in health outcomes may also contribute to economic growth. While fixed effects models help mitigate some sources of bias by controlling for unobserved heterogeneity, they do not fully eliminate issues such as reverse causality or omitted variable bias. Future research could address this limitation by employing more advanced econometric techniques, such as instrumental variable approaches.

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